Question

Write a Java program to demonstrate hybrid inheritance using class inheritance and an interface.

Steps for Execution

1. Create a package named javaprograms.
2. Create a class Number with variables a = 10 and b = 5.
3. Define an add() method in the Number class.
4. Create an interface Math containing the multiply() method.
5. Create a class Calculate that extends Number and implements Math.
6. Implement the multiply() method in the Calculate class.
7. Create a class Result that extends Calculate.
8. Define the subtract() method in the Result class.
9. Create an object r of the Result class.
10. Call add(), multiply(), and subtract() using the object.
11. Display the results.
12. Stop the program.

Algorithm

1. **Start**
2. Create superclass Number with two variables a and b.
3. Define the add() method.
4. Create interface Math with the multiply() method.
5. Create class Calculate that extends Number and implements Math.
6. Implement the multiply() method.
7. Create class Result that extends Calculate.
8. Define the subtract() method.
9. Create an object of Result.
10. Call the addition, multiplication, and subtraction methods.
11. Display the results.

12. **Stop**

UML Class Diagram

```
┌────────────────────┐
│      Number        │
├────────────────────┤
│ int a = 10         │
│ int b = 5          │
├────────────────────┤
│ + add(): void      │
└─────────┬──────────┘
          │
          │ extends
          ▼
┌────────────────────┐
│    Calculate       │
├────────────────────┤
│ + multiply(): void │
└─────────┬──────────┘
          │
          │ extends
          ▼
┌────────────────────┐
│      Result        │
├────────────────────┤
│ + subtract(): void │
└────────────────────┘
          ▲
          │
```

```
          | implements
  ┌──────────┴──────────┐
  |     Math       |
  |   <<interface>>  |
  ├─────────────────────┤
  | + multiply(): void |
  └─────────────────────┘
```

Inheritance Structure

```
Number
  |
  ▼
Calculate ◄—— implements —— Math
  |
  ▼
Result
```

This demonstrates **hybrid inheritance** by combining **multilevel inheritance** (Number → Calculate → Result) with **interface implementation** (Calculate → Math).

Code

Hybrid Inheritance Java Program

```java
package javaprograms;


class Number {
    int a = 10, b = 5;


    void add() {
        System.out.println("Addition = " + (a + b));
    }
}
```

}

```
interface Math {
    void multiply();
}

class Calculate extends Number implements Math {
    public void multiply() {
        System.out.println("Multiplication = " + (a * b));
    }
}

class Result extends Calculate {
    void subtract() {
        System.out.println("Subtraction = " + (a - b));
    }
}

public class HybridInheritance {
    public static void main(String[] args) {
        Result r = new Result();

        r.add();
        r.multiply();
        r.subtract();
    }
}
```

Output

Addition = 15

Multiplication = 50

Subtraction = 5

Result

The program successfully demonstrates **hybrid inheritance in Java** using **multilevel inheritance and an interface**.